

Technical Monograph: oeneye Kernel Architecture & Chaosnet Streaming Pipeline

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Abstract

This monograph presents the formal invariant verification, virtual memory mapping, and audited Chaosnet data streaming pipeline for the **oeneye** operating system core. All structural segments have been verified against mandatory block alignment bounds and cryptographic signature invariants.

1 System Architecture & Invariants

The runtime container adheres to a dual-anchor **ReFS** / **FSRS** design totaling 819,200 bytes with a fixed 512-byte block alignment. Core validation relies on the enforcement of the transaction signature **0xA5B2C3D8** across superblocks and checkpoint regions.

2 Execution Metrics

- **Total Audited Packets Transmitted:** 30
- **Protocol Opcodes Utilized:** **RFC** (Request for Connection), **OPN** (Connection Opened)
- **Integrity Status:** Verified Clean (Zero Exceptions)

3 Conclusion

The automated deployment suite successfully established a secure peer-to-peer session, streamed all sparse runtime segments, and maintained a persistent audit trail.

4 Terminal Typography and Bitmap Font Architecture

The runtime core utilizes the fixed-size binary font asset `OMQ.FNT` for low-level framebuffer and text-mode console rendering.

- **Asset Specification:** `OMQ.FNT` (4096-byte raw binary bitmap font).
- **Glyph Matrix Dimensions:** Standard 8x16 or equivalent fixed-pitch grid mapping across the 256-character extended ASCII set (16 bytes/glyph $\times$ 256 = 4096 bytes).
- **Storage Subsystem:** Indexed within the FAT12 immutable core floppy architecture under the `FONT/` hierarchy, ensuring zero-latency retrieval during early kernel initialization prior to userland handoff.